

Tensor products of Mycielskians of odd cycles with odd wheels are not word-representable

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Abstract

A graph is word-representable if there is a word over its vertex set, containing every vertex, in which two distinct letters alternate exactly when the corresponding vertices are adjacent. Alshammari asked whether the tensor products $\mu(C_{2n+1}) \times W_{2m+1}$ and $\mu'(C_{2n+1}) \times W_{2m+1}$ are non-word-representable for all $n \geq 1$ and $m \geq 2$, where μ and μ' denote the Mycielskian and the extended Mycielskian and W_{2m+1} is the wheel on an odd cycle. We show that the answer is yes. When $n \geq m$, each product contains an induced copy of its first factor; when $n \leq m$, both contain an induced copy of $\mu(C_{2m+1})$. These graphs are not word-representable by theorems of Hameed and of Kitaev and Pyatkin. The result, the non-word-representability of $\mu(C_{2t+1})$ and $\mu'(C_{2t+1})$ on which it relies, and the other facts its proof uses are formally verified in Lean 4. The proof was found by AI agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

All graphs are finite and simple. Two distinct letters x and y *alternate* in a word w if deleting from w all letters other than x and y leaves a word $xyxy\cdots$ or $yxyx\cdots$, of even or odd length. A graph G is *word-representable* if there is a word w over the alphabet $V(G)$ in which every vertex occurs and, for all distinct vertices x and y , the letters x and y alternate in w if and only if xy is an edge [6, 1]. Halldórsson, Kitaev and Pyatkin [4] proved that a graph is word-representable if and only if it has a semi-transitive orientation.

The *tensor product* $G \times H$ of graphs G and H has vertex set $V(G) \times V(H)$, and (a, b) is adjacent to (a', b') if and only if aa' is an edge of G and bb' is an edge of H . Kitaev and Lozin asked whether anything definite can be said about the word-representability of tensor products of two word-representable graphs, of a word-representable and a non-word-representable graph, or of two non-word-representable graphs [6, Problem 7.2.5], as quoted in [1, Problem 1.1]. Choi, Kim and Kim [2] showed that the tensor product of two graphs with semi-transitive orientations need not have one. Alshammari [1] studied tensor products of graphs containing wheels, complete graphs and (extended) Mycielskians of cycles, and posed four problems. The first of them is the following [1, Problem 5.1].

Is it true that $\mu_{2n+1} \times W_{2m+1}$ and $\mu'_{2n+1} \times W_{2m+1}$ is always non-word-representable for $n \geq 1$ and $m \geq 2$?

Here, as in [1], W_k is the wheel obtained from the cycle C_k by adding a vertex adjacent to all its vertices, $\mu_k = \mu(C_k)$ is the Mycielskian of C_k and $\mu'_k = \mu'(C_k)$ its extended Mycielskian (Section 2). We answer the question in the affirmative.

Theorem 1. *For all integers $n \geq 1$ and $m \geq 2$, the graphs $\mu_{2n+1} \times W_{2m+1}$ and $\mu'_{2n+1} \times W_{2m+1}$ are not word-representable.*

The proof (Section 3) finds a non-word-representable induced subgraph: for $n \geq m$, a copy of the first factor, embedded through a homomorphism into W_{2m+1} , as in the proof of [1, Theorem 3.6]; for $n \leq m$, a copy of μ_{2m+1} , by an embedding that adapts the one in the proof of [1, Theorem 4.3]. That these subgraphs are not word-representable is due to Hameed [5] and to Kitaev and Pyatkin [7]. We claim nothing about the other problems of [1]; in particular, whether the tensor product of two non-word-representable graphs can be word-representable [1, Problem 5.4] remains open.

Prior work. We found no earlier answer to [1, Problem 5.1]. We searched on 27 September 2026 between 18:06 and 19:02 UTC: the arXiv (its API, for the topic and the author); the works citing [1] in OpenAlex (there were none); MathDB; MathOverflow and Mathematics Stack Exchange, through the Stack Exchange API; GitHub (code, commits, issues, pull requests and repositories), including the repositories `trureturing` [10] and `SCOPE2026` [9] of AI-assisted work on open problems; Wikipedia; and web search. A second AI system repeated a smaller search later that day. A final check between 23:38 UTC on 28 September and 00:17 UTC on 29 September 2026 (the arXiv, including new versions of [1]; the works citing it in Semantic Scholar and OpenAlex; MathOverflow and Mathematics Stack Exchange; GitHub; MathDB; and web search) found nothing new. We did not search Google Scholar, Scopus or Web of Science, and we did not consult the book [6] itself. This reports the coverage of our searches, not a guarantee of priority.

2 Definitions and three known facts

For $k \geq 3$ the cycle C_k has vertex set $\mathbb{Z}_k$, with i adjacent to $i \pm 1$. The wheel W_k has the *rim* vertices r_i , $i \in \mathbb{Z}_k$, which form a copy of C_k , and the *hub* h , adjacent to every rim vertex. The Mycielskian μ_k has the *top* vertices v_i , the *shadows* u_i ($i \in \mathbb{Z}_k$) and the *root* w ; its edges are $v_i v_{i+1}$, $u_i v_{i+1}$, $u_i v_{i-1}$ and $w u_i$ for $i \in \mathbb{Z}_k$. The extended Mycielskian μ'_k has the same vertices and the edges $v_i v_{i+1}$, $u_i v_j$ for all $j \neq i$, and $w u_i$ [1, 7]. In both graphs the shadows are pairwise non-adjacent, and the root is adjacent to no top vertex.

Lemma 2.1 (well known; see [6]). *An induced subgraph of a word-representable graph is word-representable.*

Proof. If w represents G and $S \subseteq V(G)$, delete from w all letters outside S . Every letter of S still occurs, and whether two letters of S alternate depends only on the subword formed by these two letters, which does not change; so the new word represents the subgraph induced by S . $\square$

Lemma 2.2. *If $f: G \rightarrow H$ is a homomorphism of graphs, then $x \mapsto (x, f(x))$ is an isomorphism of G onto an induced subgraph of $G \times H$.*

Proof. The map is injective. For vertices x and y of G , the pair $(x, f(x)), (y, f(y))$ is an edge of $G \times H$ if and only if xy is an edge of G and $f(x)f(y)$ is an edge of H ; as f is a homomorphism, this holds if and only if xy is an edge of G . $\square$

Theorem 2 (Hameed [5]; Kitaev and Pyatkin [7, Theorem 4 and Remark 5]). *For every $t \geq 1$, the graphs μ_{2t+1} and μ'_{2t+1} are not word-representable.*

Hameed [5] proved that μ'_{2t+1} has no semi-transitive orientation; Kitaev and Pyatkin [7, Theorem 4] proved the same for μ_{2t+1} , and observed that their proof also applies to μ'_{2t+1} [7, Remark 5]. By [4], these graphs are therefore not word-representable. We use Theorem 2 only for $t \geq 2$.

3 Proof of Theorem 1

For odd integers $K \geq k \geq 3$ define the *fold* $\varphi_{K,k}: \mathbb{Z}_K \rightarrow \mathbb{Z}_k$ by $\varphi_{K,k}(i) = i$ for $0 \leq i \leq k-1$, and, for $k \leq i \leq K-1$, by $\varphi_{K,k}(i) = k-2$ if $i-k$ is even and $\varphi_{K,k}(i) = k-1$ if $i-k$ is odd. For $K = k$ it is the identity. For $K > k$ it is a homomorphism $C_K \rightarrow C_k$: the images of i and $i+1$ are adjacent for $0 \leq i \leq K-2$, and $\varphi_{K,k}(K-1) = k-1$ is adjacent to $\varphi_{K,k}(0) = 0$, as $K-1-k$ is odd. In particular $\varphi_{K,k}(i) \neq \varphi_{K,k}(i \pm 1)$ for every i .

Lemma 3.1. *Let $k \geq K \geq 5$ be odd. There are homomorphisms $\mu_k \rightarrow W_K$ and $\mu'_k \rightarrow W_K$.*

Proof. Put $\psi = \varphi_{k,K}$. Map v_i and u_i to $r_{\psi(i)}$, and w to h . The edges $v_i v_{i+1}$ and $u_i v_{i \pm 1}$ of μ_k go to the rim edges $r_{\psi(i)} r_{\psi(i+1)}$ and $r_{\psi(i)} r_{\psi(i \pm 1)}$, and $w u_i$ goes to the edge $h r_{\psi(i)}$. For μ'_k , map v_i to $r_{\psi(i)}$, every u_i to h , and w to r_0 : the edges $v_i v_{i+1}$ go to rim edges, and the edges $u_i v_j$ and $w u_i$ to edges at the hub. $\square$

Lemma 3.2. *Let $K \geq k \geq 3$ be odd, with $K \geq 5$, and let G be μ_k or μ'_k . Then μ_K is isomorphic to an induced subgraph of $G \times W_K$.*

Proof. Put $\varphi = \varphi_{K,k}$ and define ι on the vertices of μ_K by

$$\iota(v_i) = (v_{\varphi(i)}, r_i), \quad \iota(u_i) = (u_{\varphi(i)}, r_i), \quad \iota(w) = (w, h).$$

It is injective: the second coordinate determines i , or shows that the vertex is w , and the first coordinate tells top vertices from shadows.

Edges go to edges. For $v_i v_{i+1}$: the top vertices $v_{\varphi(i)}$ and $v_{\varphi(i+1)}$ are adjacent in G , as φ is a homomorphism, and $r_i r_{i+1}$ is an edge. For $u_i v_{i \pm 1}$: in μ_k the shadow $u_{\varphi(i)}$ is adjacent to $v_{\varphi(i \pm 1)}$ because $\varphi(i)$ and $\varphi(i \pm 1)$ are adjacent in C_k , and in μ'_k because they are different; and $r_i r_{i \pm 1}$ is an edge. For $w u_i$: w is adjacent to $u_{\varphi(i)}$ in G , and h to r_i .

Non-edges go to non-edges. The non-edges of μ_K are $v_i v_j$ with $j \notin \{i-1, i, i+1\}$, $u_i v_j$ with $j \notin \{i-1, i+1\}$ (including $j = i$), $u_i u_j$ with $j \neq i$, and $w v_j$. In the first two cases the second coordinates r_i and r_j are not adjacent in W_K , as $j \notin \{i-1, i+1\}$ and W_K has no loops. For $u_i u_j$ the first coordinates are two shadows, or one shadow twice, which are not adjacent in G ; for $w v_j$ the first coordinates w and $v_{\varphi(j)}$ are not adjacent in G . So ι is an isomorphism onto the subgraph of $G \times W_K$ induced by its image. $\square$

Proof of Theorem 1. Let $k = 2n + 1$ and $K = 2m + 1 \geq 5$, and let G be μ_k or μ'_k . If $n \geq m$, Lemmas 3.1 and 2.2 give an induced copy of G in $G \times W_K$. If $n \leq m$, Lemma 3.2 gives an induced copy of μ_K in $G \times W_K$. In both cases the copy is not word-representable by Theorem 2, since $k \geq K \geq 5$ in the first case; so $G \times W_K$ is not word-representable by Lemma 2.1. $\square$

4 Formal verification

The results of this paper are formally verified in Lean 4 (version 4.34.1) with Mathlib [8] (version v4.34.1, commit d13f23b7).

Statements before proofs. After the proof had been found, the agent that found it wrote a Lean file of definitions and statements, the *statement file*, which contains no proofs. An independent agent reviewed version 1 of this file against the sources and tested its definitions with programs of its own; version 2 applies its corrections and adds the lemmas used in the proof. A second AI system reviewed version 2, and it was frozen on 27 September 2026 at 22:04 UTC by recording its SHA-256 hash

40c8efd69f8499c9874f07163043c0793c94c8597b7a3902d5aeb0a2ce0c9b58.

The proofs were written after the freeze, in separate files that import the statement file, which was not changed.

What is proved. Table 1 lists the eleven statements of the frozen file; the file of checks proves each of them. In ASCII notation, the definition of word-representability and the statement of Theorem 1 for μ read as follows (implicit arguments omitted), with `forall`, `exists`, `->`, `<->`, `/\`, `in`, `!=` and `Not` for $\forall$, $\exists$, $\rightarrow$, $\leftrightarrow$, $\wedge$, $\in$, $\neq$ and $\neg$.

```
def Alternate (w : List V) (x y : V) : Prop :=
  noRepeat (w.filter (fun z => z == x || z == y)) = true
def Represents (G : SimpleGraph V) (w : List V) : Prop :=
  (forall v : V, v in w) /\
  forall x y : V, x != y -> (G.Adj x y <-> Alternate w x y)
def WordRepresentable (G : SimpleGraph V) : Prop := exists w : List V, Represents G w
def Problem1Mu : Prop := forall n m : Nat, 1 <= n -> 2 <= m ->
  Not (WordRepresentable (tensorProd (mu (2 * n + 1)) (wheel (2 * m + 1))))
```

Here `noRepeat` says that no two consecutive letters are equal, and `mu n` and `muExt n` stand for `mycielskian (cycleGraph n)` and `extMycielskian (cycleGraph n)`. The name `Problem1Mu` comes from the label of [1, Problem 5.1] in the source of [1]. Lemmas 2.1, 2.2, 3.1 and 3.2 are proved in Lean as in the text.

Lean statement	what it says	here
<code>Problem1Mu</code> , <code>Problem1MuExt</code>	$\mu_{2n+1} \times W_{2m+1}$, resp. $\mu'_{2n+1} \times W_{2m+1}$, is not word-representable for all $n \geq 1, m \geq 2$	Thm 1
<code>Problem5_1</code>	the conjunction of the two	Thm 1
<code>MycielskiOddCycleNotWR</code> , <code>ExtMycielskiOddCycleNotWR</code>	μ_{2t+1} , resp. μ'_{2t+1} , is not word-representable for all $t \geq 1$	Thm 2
<code>WordRepresentableOfEmbedding</code>	a graph that embeds as an induced subgraph into a word-representable graph is word-representable	Lem. 2.1
<code>EmbeddingOfHom</code>	a homomorphism $G \rightarrow H$ gives an embedding of G into $G \times H$ as an induced subgraph	Lem. 2.2
<code>Lemma1Mu</code> , <code>Lemma1MuExt</code>	a homomorphism $\mu_{2n+1} \rightarrow W_{2m+1}$, resp. $\mu'_{2n+1} \rightarrow W_{2m+1}$, for $2 \leq m \leq n$	Lem. 3.1
<code>Lemma2Mu</code> , <code>Lemma2MuExt</code>	an embedding of μ_{2m+1} as an induced subgraph into $\mu_{2n+1} \times W_{2m+1}$, resp. $\mu'_{2n+1} \times W_{2m+1}$, for $1 \leq n \leq m, m \geq 2$	Lem. 3.2

Table 1: The eleven statements of the frozen statement file (namespace `WordRepTensor.Challenge`). Each is proved.

How Theorem 2 is proved in Lean. The Lean proof does not follow [5, 7]; it works with words directly. Let $x \neq y$ be letters of a word, and let x occur first. Then x and y alternate if and only if every prefix of the word contains at least as many letters x as letters y , and at most one more. Now let a word represent a graph, and orient each edge from the end vertex that occurs first. Write $\#x$ for the number of letters x in a prefix. If $a \rightarrow c \rightarrow d \rightarrow b$ is a directed path and a is adjacent to b , then a occurs before b , so in every prefix each of $\alpha = \#a - \#c$, $\beta = \#c - \#d$, $\gamma = \#d - \#b$ and $\delta = \#a - \#b$ is 0 or 1. Hence $\#a - \#d = \alpha + \beta = \delta - \gamma$ and $\#c - \#b = \beta + \gamma = \delta - \alpha$ are 0 or 1; as a occurs before d , and c before b , the vertex a is adjacent to d , and c to b . For μ_{2t+1} and μ'_{2t+1} with $t \geq 2$, consider the seven vertices $w, v_i, v_{i+1}, v_{i+2}, u_i, u_{i+1}, u_{i+2}$. A computation checked by the Lean kernel runs through all orientations of the edges among them (2^9 for μ , 2^{11} for μ') and shows that in every orientation with the property just proved, the orientation of the five edges between the columns i and $i+1$ (the edges $v_i v_{i+1}$, $v_i u_{i+1}$, $u_i v_{i+1}$, $w u_i$ and $w u_{i+1}$) and that of the five corresponding edges between the columns $i+1$ and $i+2$ get different colours, for a fixed colouring of the 2^5 orientations of five edges with two colours. Going once around the odd cycle changes the colour an odd number of times, which is impossible. The graphs μ_3 and μ'_3 are equal, and a computation over the 2^{12} orientations of their twelve edges shows that none has the property.

Checks. For every statement of Table 1, the axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`. A static scan found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern`, `unsafe` or `opaque`, and no axiom declaration, in the proof files. The Lean package of this paper, 11 modules (the statement file, its tests, eight proof modules and the file of checks), was built module by module in an empty directory, reusing only the pinned toolchain and the compiled Mathlib and its dependencies, and every module was checked with `leanchecker`, all with exit code 0, twice (on 28 September 2026 between 20:12 and 20:21 UTC). For the release, the whole Lean project of the repository was rebuilt from a clean directory and all its modules were checked again with `leanchecker` (28 September 2026, 20:48–22:16 UTC). The Lean files are in version 1.9.0 of the repository [3], directory `lean/Research/WordRepTensor/`.

Definitions the reader must trust. Apart from Mathlib’s graphs, cycle graphs (`cycleGraph`), homomorphisms, embeddings and lists, the statements use the definitions of alternation, representation and word-representability shown above; the tensor product (`tensorProd`, built with Mathlib’s `SimpleGraph.fromRel`); the wheel (`wheel`, on `Sum (Fin n) Unit`); and the Mycielskian and the extended Mycielskian (`mycielskian`, `extMycielskian`, on `Sum V (Sum V Unit)`), with top vertices, shadows and root in this order). A file of known-answer tests checks these definitions on small graphs: adjacencies and edge counts of cycles, wheels, Mycielskians and tensor products; words that represent C_4 , C_5 , K_3 and $K_2 \times K_2$; words that do not represent a given graph, including words that omit a vertex; small instances of the maps of Lemmas 2.2, 3.1 and 3.2; and a wrong map that is not an embedding.

Faithfulness and non-claims. A word is a list of vertices in which every vertex must occur, as in [1]; so the empty word represents no graph with a vertex. The Lean statements concern word-representability itself, not semi-transitive orientations, so the theorem of [4] is not needed as a hypothesis. Theorem 2 is formalized in its word-representability form; we do not formalize the characterizations of [5, 7] for arbitrary graphs. That the Lean definitions agree with those of [1] was checked by the reviewing agents, not by a machine.

How this paper was produced

This work was carried out from 27 to 29 September 2026 (UTC) in a workflow directed by the author, using AI agents. Agents of Anthropic Claude Code, with the model Claude Opus 5.5, selected the question, found the proof, checked it with two separately written programs for small parameters, wrote the statement file of Section 4, carried out the main literature search described in the introduction, and drafted this text. An independent Claude Code agent reviewed version 1 of the statement file against the sources and checked the definitions with its own programs; its corrections were applied in version 2. OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) then checked the mathematics independently, recomputing the embeddings with its own program, reviewed version 2 of the statement file and froze it. When Codex’s usage quota ran out, Claude Code wrote the Lean proofs and ran the checks of Section 4, and an independent Claude Code agent reviewed the proofs. Another independent Claude Code agent refereed a draft of this paper. Codex has not reviewed the Lean proofs or this paper; the final review before release was done by an independent Claude Code agent in its place. Throughout, “we” refers to this workflow. The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the result rests on the Lean proofs of the statements in Table 1 and on the agreement of those statements with the definitions of [1], which was checked by the reviewing

agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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References

- [1] N. Alshammari, On the word-representability of tensor product graphs, arXiv:2609.20881v1, 16 September 2026; we use its numbering.
- [2] I. Choi, J. Kim and M. Kim, On operations preserving semi-transitive orientability of graphs, *J. Comb. Optim.* **37** (2019), 1351–1366, <https://doi.org/10.1007/s10878-018-0358-7>; arXiv:1810.11620.
- [3] H. Dong, *ai4math-results*, repository, <https://github.com/Horace-Maxwell/ai4math-results>, version 1.9.0 (2026); archived at Zenodo under the concept DOI <https://doi.org/10.5281/zenodo.22970254>.
- [4] M. M. Halldórsson, S. Kitaev and A. Pyatkin, Semi-transitive orientations and word-representable graphs, *Discrete Appl. Math.* **201** (2016), 164–171, <https://doi.org/10.1016/j.dam.2015.07.033>.
- [5] H. Hameed, On semi-transitivity of (extended) Mycielski graphs, *Discrete Appl. Math.* **359** (2024), 83–88, <https://doi.org/10.1016/j.dam.2024.07.028>.
- [6] S. Kitaev and V. Lozin, *Words and Graphs*, Monographs in Theoretical Computer Science, an EATCS Series, Springer, 2015, <https://doi.org/10.1007/978-3-319-25859-1>.
- [7] S. Kitaev and A. Pyatkin, A note on Hameed’s conjecture on the semi-transitivity of Mycielski graphs, *Discuss. Math. Graph Theory* **45** (2025), <https://doi.org/10.7151/dmgt.2575>; quoted from arXiv:2408.05066v1 (“A note on semi-transitivity of Mycielski graphs”), whose numbering we use.
- [8] The mathlib Community, The Lean mathematical library, in *Proc. 9th ACM SIGPLAN Int. Conf. Certified Programs and Proofs* (CPP ’20), ACM, 2020, <https://doi.org/10.1145/3372885.3373824>.
- [9] SCOPE-Science, SCOPE2026 repository, <https://github.com/SCOPE-Science/SCOPE2026>, accessed 27 September 2026.
- [10] The Omega Institute, *trureturing* repository, <https://github.com/the-omega-institute/trureturing>, accessed 27 September 2026.